Psychometrics - Data Assignment 2

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Introduction

Life satisfaction is a vital measure of well-being and can reflect an individuals overall view of their quality of life. This report aims to investigate changes in Life Satisfaction among young adults across two time point by conducting longitudinal confirmatory factor analysis using the framework of Structural Equation Modeling (SEM). The dataset consists of responses gathered from a large-scale survey that was administered in 1987 and 1991. The study has 1,775 participants with a focus on individuals aged 17 to 27 in 1987. The Life Satisfaction construct was evaluated by a series of Likert-scaled indicators that measured participants' perceptions of their overall life circumstances. In this context, SEM is used to evaluate the Life Satisfaction construct over time whilst testing for measurement invariance. The main objective of this analysis is to determine whether Life Satisfaction improved or deteriorated between 1987 and 1991, while ensuring that the construct was measured consistently over time.

Methods

Dataset

As mentioned, the dataset contains responses from 1,775 young adults, including a variety of psychological and demographic variables. This study utilizes two repeated measurements collected from the same individuals in 1987 and 1991. This analysis focuses specifically on three indicators that were used to measure the latent variable, Life Satisfaction, for each time point.

For both 1987 and 1991, participants responded to three statements, each on a 7-point Likert scale - with higher values indicating greater satisfaction.

1987 Indicators	1991 Indicators	Statements
S341	SS281	In most respects my life is ideal
S342	SS282	My life conditions are excellent
S343	SS283	Overall I am satisfied with my life

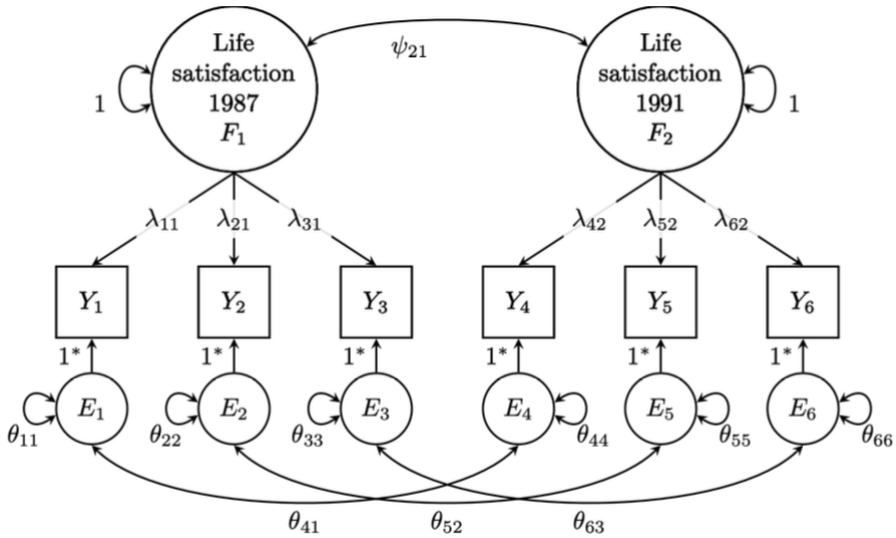
Statistical Analysis

Data Preparation

The data contained missing values for the indicators which were assumed to follow a Missing Completely at Random (MCAR) mechanism. To address this, the missing data was removed using listwise deletion. This resulted in the removal of 530 individuals, leaving a final sample of 1,245 participants for analysis. The Likert-scale responses for all indicators were assumed to be numerical. Following data preparation, descriptive statistics (means, standard deviations, covariances) were computed for the indicators.

Identification

Before conducting our analysis, it is important to check if the model is identified or not. The concept of identification in SEM refers to whether enough information is available in the data to estimate the model parameters uniquely. A model is considered identified if the degrees of freedom are positive, meaning there must be more known data points (i.e., variances and covariances) than parameters to estimate.



$$df = 1/2(p(p+1)) - q$$

In our data, the total estimated parameters are:

$$q = 6(\text{factor loadings}) + 1(\text{latent factor covariance}) + 3(\text{error covariances}) + 6(\text{observed variances}) = 16$$

Hence, the degrees of freedom are:

$$df = 1/2(6(6+1) - 15 = 21 - 16 = 5$$

The degrees of freedom are indeed positive, indicating the model is over-identified. There is more known information than unknown information. The model can be statistically evaluated, so we continue the analysis by testing for measurement invariance.

Longitudinal Factor Analysis

This study employs SEM which allows for the simultaneous estimation of relationships between latent constructs and their observed indicators while accounting for measurement error. This will be done by conducting longitudinal factor analysis using the **lavaan** package in **R**. Fitting a longitudinal factor model is not very different from fitting a factor model with means, but it includes covariances between the residual factors of the same indicator at different time points. The primary focus is to test for longitudinal measurement invariance to see if we can make latent mean comparisons over time.

The three levels of invariance that were tested are:

1. **Configural Invariance**: This is the first level of invariance that we need to test. Configural invariance assesses whether the same **factor structure** (i.e., the same number of factors and the same pattern of loadings) holds across both time points without constraining any parameters. The factor loadings, intercepts, and error variances are allowed to vary across time points.

2. **Weak Factorial Invariance (Metric Invariance)**: This model constrains the **factor loadings** to be equal across time, testing whether the relationships between the observed items and the latent construct Life Satisfaction are consistent between 1987 and 1991. With this equality constraint the variance of Life Satisfaction for the second time point is already identified, therefore, we can freely estimate this parameter. Moreover, we allow intercepts to vary.

3. **Strong Factorial Invariance (Scalar Invariance)**: This model further constrains the **intercepts** of the observed items to be equal across time, testing whether the baseline levels of the latent construct are comparable across the two time points. This allows us to freely estimate the common factor mean of Life Satisfaction at the second time point.

Indicators such as the Chi-Square test, Root Mean Square Error of Approximation (RMSEA), and Comparative Fit Index (CFI) were used to evaluate model fit at each stage of the analysis. Additionally, a Chi-Square Difference Test was conducted to compare models at different levels of invariance. The invariance models were tested consecutively, with each model imposing progressively stricter constraints on the parameters as detailed above. By testing these models sequentially, the analysis can assess changes in the latent variable (Life Satisfaction) more accurately.

Results

Descriptive Statistics

The following three tables illustrate the mean, standard deviation and covariance for each indicator. These descriptives provide an initial insight into the distribution of Life Satisfaction responses and the relationships between the indicators over time, which will be further explored in the confirmatory factor analysis.

	Mean
S341	5.072289
S342	5.502008
S343	5.874699
SS281	5.225703
SS282	5.606426
SS283	5.853815

The mean values for the 1987 indicators range from 5.07 to 5.87. In 1991, the means are similarly distributed, with values ranging from 5.23 to 5.85. The values suggest relatively high levels of Life Satisfaction across both time points.

	Standard Deviation
S341	1.411794
S342	1.307671
S343	1.248296
SS281	1.256359
SS282	1.204610
SS283	1.185861

The standard deviations for the 1987 indicators range from 1.25 to 1.41, while the 1991 indicators have slightly lower variability, with standard deviations between 1.18 and 1.26. This indicates that, although there is some variability in responses, most participants provided responses around the mean values.

Table 4: Covariance Matrix

	S341	S342	S343	SS281	SS282	SS283
S341	1.9931624	1.1461570	1.0372002	0.5785263	0.5019467	0.4382288
S342	1.1461570	1.7100040	0.8981618	0.4058969	0.5353569	0.3877536
S343	1.0372002	0.8981618	1.5582420	0.3466315	0.3919634	0.4430829
SS281	0.5785263	0.4058969	0.3466315	1.5784385	0.9916354	0.9767507
SS282	0.5019467	0.5353569	0.3919634	0.9916354	1.4510841	0.8515832
SS283	0.4382288	0.3877536	0.4430829	0.9767507	0.8515832	1.4062656

The covariance values are positive, suggesting a positive association between the Life Satisfaction indicators from 1987 and 1991.

From initial exploration of the data, it seems that Life Satisfaction did slightly improve during the span of these four years. However, simply comparing these measurements of each indicator means that we assume the indicators are perfect reflections of the latent construct (Life Satisfaction), which is rarely the case. Therefore, we need to perform more rigorous analyses to account for measurement error and test for measurement invariance in order to make direct comparisons of the latent construct at different time points.

1. Configural Invariance

The lavaan function returned the following fit measures for the configural invariance model:

Table 5: Configural Invariance Model

Measure	Value
Chi-squared	9.146
df	5.000
p-value	0.103
RMSEA	0.026
CFI	0.999

According to the chi-squared test, we fail to reject the null hypothesis since the p-value is greater than 0.05. This suggets that the configural invariance model fits the data well. The RMSEA measures the extent to which a model approximates the data per degree of freedom. An RMSEA value of 0.05 or lower generally indicates a good fit. In this case, RMSEA = 0.026, which is well within the range of excellent fit. The CFI compares the fit of the model to the fit of an independent model, where no relationships between the variables are assumed. A CFI value above 0.95 indicates excellent fit. In this case, CFI = 0.999, suggesting that the model provides a very good fit to the data. Overall, the configural invariance model seems to fit the data well when considering all three fit measures, hence we can move onto the next level and set more constraints.

2. Weak Factorial Invariance

We now constrain the factor loadings to be equal across the two time points. This ensures that the relationship between the latent factor (Life Satisfaction) and the observed indicators is consistent overtime.

The results of the weak factorial invariance model are depicted in the table below.

Table 6: Weak Factorial Invariance Model

Measure	Value
Chi-squared	13.762
df	7.000
p-value	0.056
RMSEA	0.028
CFI	0.998

The chi-squared test returned a p-value greater than 0.05, hence we do not reject the null hypothesis. This suggets that the weak factorial invariance model fits the data well. This is also supported by the RMSEA and CFI values.

The fit of the weak factorial invariance model can be compared to the fit of the configural invariance model. This is to test whether the equality constraints on the factor loadings lead to significant deterioration in model fit. To check this, we perform an analysis of variance (ANOVA) test.

When the deterioration in fit is not significant, we may conclude that the assumption of weak factorial invariance holds.

$$H_0 : \text{There is no significant difference in how the models fit the data.}$$

$$H_A : \text{There exists a significant difference in how the models fit the data.}$$

Model	AIC	BIC	Chi.Squared.Difference	P.Value
Configural	21756	21854		
Weak	21756	21858	4.6163	0.09944

According to the ANOVA results, we do not reject the null hypothesis. This indicates that the models are not significantly different and both fit the data well. This is supported by the AIC and BIC values. Therefore, we choose the less complex model (weak factorial invariance) that has fewer parameters to estimate by the principle of parsimony. Moreover, this means that constraining the factor loadings to be equal across time does not worsen the model fit. Since the deterioration in fit is not significant, we may say that the assumption of weak factorial invariance holds, and test the next level.

3. Strong Factorial Invariance

The results of the strong factorial invariance model are depicted in the table below.

Table 8: Strong Factorial Invariance Model

Measure	Value
Chi-squared	29.918
df	9.000
p-value	0.000
RMSEA	0.043
CFI	0.993

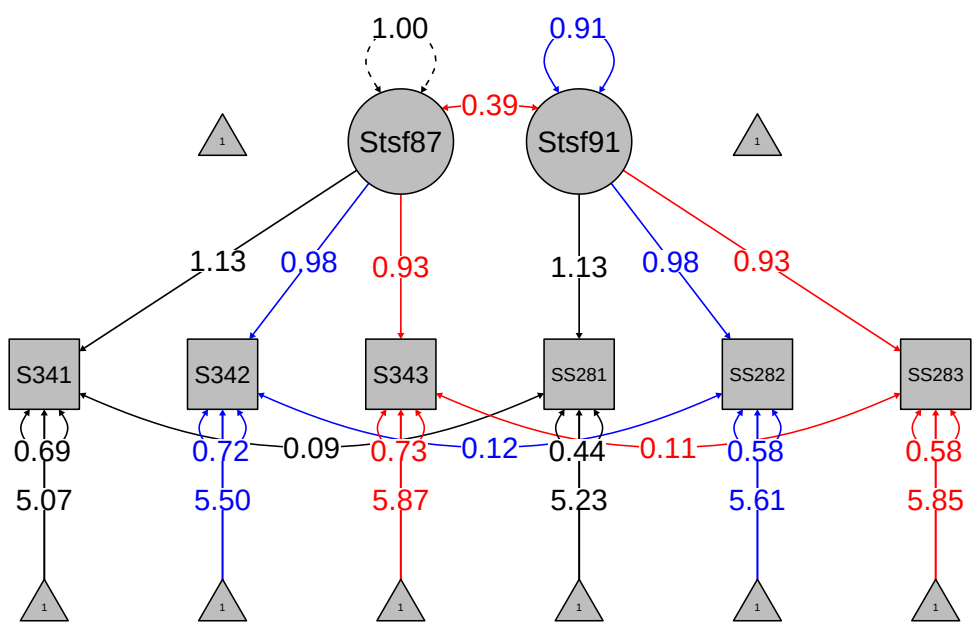
The chi-square test has a p-value of 0, indicating that the model does not fit the data perfectly. However, this is contradicted by the CFI and RMSEA values which indicate a relatively good fit. The chi-squared test is sensitive to large sample sizes. Hence given the CFI and RMSEA we can say that the strong factorial invariance model provides an acceptable fit to the data.

To test whether the assumption of strong factorial invariance holds, we can compare this model to the weak factorial invariance model.

Model	AIC	BIC	Chi.Squared.Difference	P.Value
Weak	21756	21858		
Strong	21768	21860	16.156	0

The p-value is less than 0.05, which means we reject the null hypothesis that. This indicates that the models are significantly different and further implies that deterioration in fit is significant. Thus, suggesting that the equality constraint on the intercepts (i.e., imposing strong invariance) significantly worsens the model. Therefore, strong factorial invariance does not hold. Moreover, this means that the observed items (S341, S342, S342, etc.) do not have the same baseline (intercepts) across time points (1987 and 1991).

Final Model



The above factor model illustrates the weak factorial invariance model - our final model. We can see that the factor structure is the same and the factor loadings are equal for respective indicators over time. For example, S341 and SS281 represent the same statement and have a factor loading of 1.13.The covariance between Life Satisfaction in 1987 and in 1991 is 0.39. However, the intercepts for each indicator vary.

Discussion

From the results of our analysis, our final model is the weak invariance factorial model. This tells us that the structure of how Life Satisfaction is measured over time is stable. Additionally, we can confidently assume that the relationships between the latent construct, Life Satisfaction, and its indicators is consistent over time. This means that the contribution of the items in measuring Life Satisfaction is the same in both years. The covariance between the latent variables (Life Satisfaction 1987 and Life Satisfaction 1991) is estimated to be 0.39 which suggests that there is a positive relationship between Life Satisfaction scores at the two time points. However, this information does not imply that Life Satisfaction increased or decreased over time.

The lack of strong factorial invariance introduces implications for interpreting true changes in Life Satisfaction. Since the intercepts are not equal across the two time points, this suggets potential bias in how individuals responded to the items over time. Factors such as major life events (marriage, job loss, health issues), shifts in personal standards, or societal changes may have influenced how participants perceive Life Satisfaction during these four years. This variation in baseline levels makes it difficult to compare latent means over time. Without strong factorial invariance, we cannot make reliable interpretations about the improvement or deterioration in Life Satisfaction from 1987 to 1991. Therefore, we are not able to make valid conclusions about changes in Life Satisfaction levels over time.